RS-2 Javelin

Specifications and assembly instructions

Description

The RS-2 Javelin stage is a high tech upper stage for orbital launches. The stage has its on board computer and is designed to insert payloads into low kerbin orbit. Once a payload is inserted, the stage can return to Kerbin on its own and be recovered using its parachutes.

Basic construction

Construction is best performed using two subassemblies. First, an avionics bay is built which houses computer, parachutes and batteries. Once this subassembly is built, the rest of the stage is mostly stacked axially.

There are two RS-2 variants: a regular one and an extended range version with an extra 100 unit LFOX tank. An interim variant with an OKTO computer can be built. These downgraded variants have the O suffix added to its model name.

Variant	Fuel amount
RS-2A	400
RS-2B (aka RS-2 extended range)	500

Command system

The stage has its own HECS on-board computer and redundant radios that enable it to fully control the launch process up to orbital insertion.

Guidance system

Guidance is provided by engine gimballing and assisted somewhat by the computer reaction wheel. Payload assistance is possible.

Recovery system

Stage recovery is provided by the parachutes. Full control is possible and hybrid parachute/propulsive landings can be achieved. Pure propulsive landings are not possible due to engine inefficiencies at sea level.

Assembly instructions

Use of prefabricated subassemblies is recommended but not necessary. Always store a set of savefiles for each variant.

Full bill of materials

1x	Probodobodyne HECS (OKTO in downgraded models)
1x	Z-200 Rechargeable Battery pack
1x	Mk16 Parachute
1x	FL-T400 Fuel Tank
1x	FL-T100 Fuel Tank (ER model only)
2x	Communotron 16-S
1x	LV-909 "Terrier" Liquid Fuel Engine
1x	Service Bay (1.25m)

Technology

The stage requires a technology level 5 with Precision Engineering (level 6) added. Derated versions of the stage can be created with pure level 5 technology.

- Precision Engineering
  - Probodobodyne HECS
- Electrics
  - Z-200 Rechargeable Battery pack
- Survivability
  - Mk16 Parachute
  - Service Bay (1.25m)
- Advanced Rocketry
  - FL-T400 Fuel Tank
  - LV-909 "Terrier" Liquid Fuel Engine
- Basic Rocketry
  - FL-T100 Fuel Tank (ER model only)
- Engineering 101
  - Communotron 16-S

Avionics bay

Parts list

1x	Probodobodyne HECS (OKTO on downgraded models)
1x	Z-200 Rechargeable Battery pack
1x	Mk16 Parachute
1x	Service Bay (1.25m)

Assembly procedure

Start with OKTO or HECS computer

Set Hibernate in warp to AUTO

Install Z-200 battery on axially on top of computer

Install parachute axially on top of battery

Set parachute parameters

Min pressure	0.45
Altitude	1250
Spread angle	0

Install payload bay so that lower part of computer is installed axially at bottom of bay

Match HECS/OKTO label to service bay door (E orientation)

Spare parts 0

Parameters

Part count: 4

Subassembly cost: 1942 (OKTO), (HECS)

Subassembly mass: 310 Kg (OKTO), Kg (HECS)

Final Assembly

Parts list

1x	Avionics bay
1x	FL-T400 Fuel Tank
1x	FL-T100 Fuel Tank (ER model only)
2x	Communotron 16-S
1x	LV-909 "Terrier" Liquid Fuel Engine

Assembly procedure

Start with T400 tank.

Use dark "checkerboard" pattern.

Install communotron antennas radially on tank.

2x symmetry with snap

Vertical orientation

Antenna should barely touch "rivet ring" panel on top and checkerboard panel (but not checkerboard) at the bottom.

E/W orientation

Install LV-909 axially under tank.

On ER model, install T100 tank on top of T400 tank

Install avionics bay axially on top of uppermost tank

Doors oriented E/W

Re-root to service bay

Figure 1: Top view of RS-2A showing antenna placing

Stage configuration

Part	Variable	Value	Default	Notes
Parachute	Pressure	0.45	No	
	Altitude	1250	No	
	Angle	0	No	
	Mode	Safe	Yes	
Service bay	Spare parts	0	No	
	Door	Closed	Yes	Close after inspecting items inside
Z-200	Electric charge	FULL	Yes	
Computer	Vessel naming	SET	No	
	Hibernation	Off	Yes	
	Hibernate in warp	Auto	No	
	Control point	Default	Yes	
	Reaction wheel	Normal/active	Yes	
T100 tank	LFOX	FULL	Yes	B model only
	Flow priority	-1	No	B model only
T400 tank	LFOX	FULL	Yes	
	Flow priority	0	Yes	
Radios	Retract/deploy	Active	Yes	
	Deactivate threshold	20	Yes	
	Activate threshold	80	Yes	
Engine	Throttle	Main	Yes	
	Thrust limiter	100	Yes	
	Gimbal	100/Free	Yes	

Staging

Stage	Parts	Notes
0	Parachutes	
1	Engine	

Parameters

Model	A	AO	B	BO
Part count	8	8	9	9
Cost (full)		3432		3582
Cost (dry)		3248		3353
Mass (full)		3090		3653
Mass (dry)		1090		1153

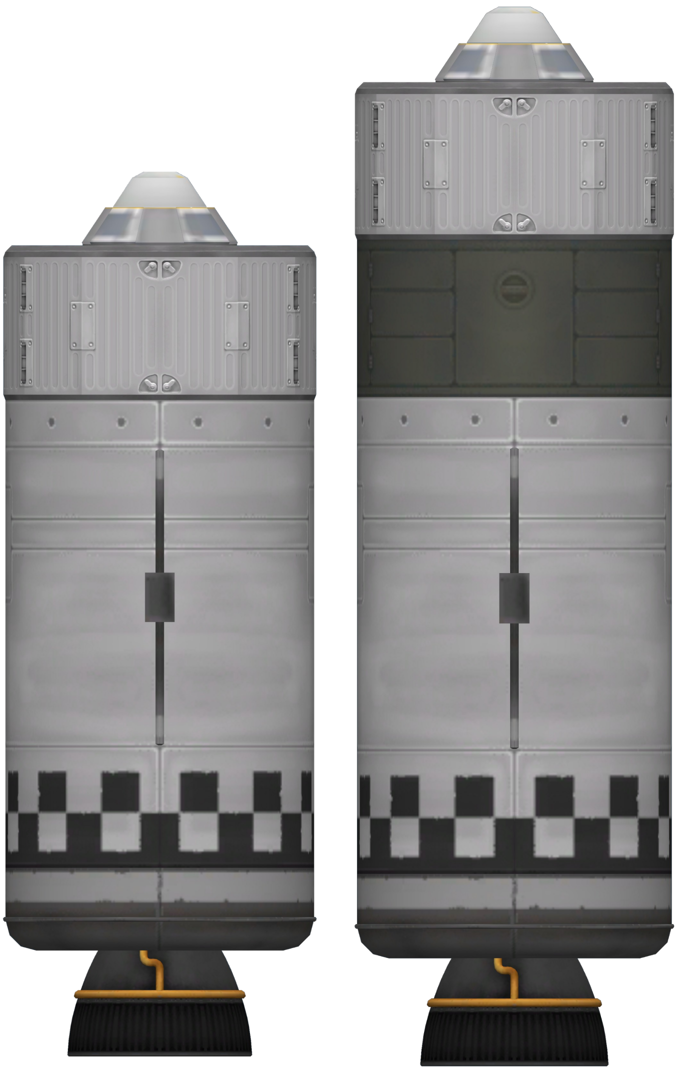

Figure 2: RS-2 comparison, A model to the left, B model to the right